(i) **Neuron or nerve cell:** A neuron or nerve cell is the structural and functional unit of the nervous system. It is the longest cell of the human body with a length of over 100µm. These cells are highly specialised to detect, receive and transmit different kinds of stimuli. Information is conducted through neurons in the form of electrical impulses from one part of the body to another.

(ii) **Neuroglia**: Neuroglia are also called as **glial cells.** They are non-exciting, supporting cell of the nervous system. They do not initiate or conduct nerve impulses.

(iii) **Nerve fibres:** The nerve fibres are the long slender processes of neurons. A number of nerve fibres are bundled up together to form **nerves**.

15.1.1 Structure of Neuron

A neuron typically consists of three basic parts: Cyton, Dendrites and Axon.

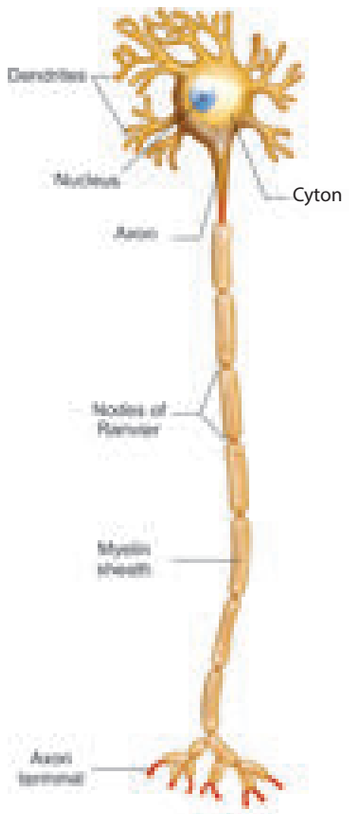


Fig. 15.1 Structure of Neuron

(i) **Cyton**: Cyton is also called cell body or perikaryon. It has a central nucleus with abundant cytoplasm called **neuroplasm**. The cytoplasm has large granular body called **Nissl's granules** and the other cell organelles like mitochondria, ribosomes, lysosomes, and endoplasmic recticulum. Neurons do not have the ability to divide. Several neurofibrils are present in the cytoplasm that help in transmission of nerve impulses to and from the cell body.

(ii) **Dendrites**: These are the numerous branched cytoplasmic processes that project from the surface of the cell body. They conduct nerve impulses towards the cyton. The branched projections increase the surface area for receiving the signals from other nerve cells.

(iii) **Axon**: The axon is a single, elongated, slender projection. The end of axon terminates as fine branches which terminate into knob like swellings called **synaptic knob**. The plasma membrane of axon is called **axolemma,** while the cytoplasm is called **axoplasm.** It carries impulses away from the cyton. The axons may be covered by a protective sheath called **myelin sheath** which is further covered by a layer of **Schwann cells** called **neurilemma**. Myelin sheath breaks at intervals by depressions called **Nodes of Ranvier**. The region between the nodes is called as **internode**. Myelin sheath acts as insulator and ensures rapid transmission of nerve impulses.

Synapse: A junction between synaptic knob of axon of one neuron and dendron of next neuron is called **synaptic junction**. Information from one neuron can pass to another neuron through these junctions with the release of chemicals known as neurotransmitters from the synaptic knob.

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